

Technical Presentation

Ngô Minh Nhựt

2024

Why we need to give a good presentation

- ❑ First impression matter
- ❑ Chances to explain what we do. There is no point doing good work if others cannot understand what we did
- ❑ Good practice for a teaching career. Good practice for any career
- ❑ Help sort out what we have done and understand it better ourselves



ám chỉ tới nghiên cứu
có câu chuyện, thể hiện được sự am hiểu.

Types of presentations

- ❑ Quick 1-minute “what I do” talk dành cho nghiên cứu sinh
- ❑ 25-minute conference paper presentation
- ❑ Project presentation
- ❑ Thesis defense
- ❑ Job talk

Types of presentations

What they have in common

- ❑ Never enough time to talk about everything
- ❑ All of them reflect on you and need practice/polish
- ❑ Focus on a clear goal and message

có mở bài, thân bài, kết bài (1 câu chuyện đầy đủ, điều chỉnh theo thời gian)

có bản raw thuyết trình

có số trang (quan trọng), không đọc những gì có trên slide

****bài của tôi có phần hay ở chỗ này nè, hạn chế chỗ này nè ****

Some good tips

- ❑ Be brief ngắn gọn => trọng tâm (slide) => giải thích (người nói)
 - Use keywords rather than long sentences
- ❑ Avoid cramming too much into one slide
 - Do not be a slave to your slides
- ❑ Avoid covering up slides
 - If time is enough, do not rush the slides

Some good tips

- ❑ Use **large** fonts: make text easy to read from the back of the room
- ❑ Use **colors** to emphasize
- ❑ Use illustrations to get across key concepts
 - Images and visualizations are helpful to convey information
 - Include diagrams to show how your system works
 - May include limited animation

One picture is worth a thousand words

Some good tips

- ❑ Make eye contact 
- ❑ Be ready to skip slides if time is running out
- ❑ Practice
 - Give at least one practice talk before presenting in front of audiences
 - Practice the talk to yourself, e.g., speaking out loud in front of a mirror
- ❑ Be neat
 - Be in a dress you like the most. It will make you feel comfortable

Typical outline for project talk

- ❑ Title/author/affiliation (1 slide)
- ❑ Introduction (1 slide)
- ❑ Outline (1 slide)
- ❑ Background (2-6 slides)
- ❑ Results (2-6 slides)
- ❑ Summary/Conclusion (1-2 slides)

Typical outline for project talk

- Title/author/affiliation (1 slide)
 - Use descriptive slide titles
 - Choose a descriptive title that helps the audience to appreciate what the specific contribution of this slide is

Typical outline for project talk

□ Introduction (1 slide)

- Start your talk with motivation and examples of the problem you are solving
- Convince the audience that this talk is worth paying attention to

Typical outline for project talk

□ Outline (1 slide)

- Can be useful, especially in a talk that runs longer than 30 minutes
- Present an outline slide at the beginning of each major section
 - Indicate current section with color, font, and/or an arrow
 - Help remind your audience on the structure of your presentation

Outline

- Problem statement
- **Proposed architecture** ⇐
- Experiments

Typical outline for project talk

□ Background (2-6 slides)

- Motivation
- Related work
- Methods

Why is it important?

What have others done?

What is my approach?

Typical outline for project talk

- Results (2-6 slides)
 - Present key results and key insights
 - Do not try to show all results
 - Use tables and charts

Typical outline for project talk

- Summary/Conclusion (1-2 slides)
 - Remind audience of take-home messages of the talk
 - 3 things to remember
 - Leave your conclusion/contribution slide up while you are answering questions.

Other things to consider

- ❑ Oral communication is different from written communication
 - Keep it simple and focus on a few key points
 - Repeat key insights

Other things to consider

- ❑ Manage time: better to give a complete story than telling everything
 - Do rehearsal and keep track how long you present
 - Do not be afraid of removing slides if it took too long
 - **Number of slides** should be **equal to number of minutes** you would have

Other things to consider

- ❑ Include slide numbers. Audiences will spend less time trying to describe what slide their comment applies to
- ❑ Be sensitive to your audience
 - The same talk may need to be adjusted for a different audience
- ❑ Make audience want to learn more
- ❑ Handle Q&A as important as the format talk itself

Common mistakes

❑ Use too small size text

- This is text that you cannot read

If something isn't important enough for your audience to be able to read, then it probably does not belong on your slides.

❑ Ignoring visual design principles

- ✗ Overcrowding slides with text or using illegible fonts and poor color contrast.
- ✓ Keep slides visually clean with bullet points, high-quality images, and readable fonts.
- ✓ Use visuals to complement, not overwhelm.

Common mistakes

❑ Poor time management

- ✗ Spending too much time on certain sections, leading to a rushed conclusion, your important parts still not presented, or running out of time.
- ✓ Practice timing and allocate specific time slots for each section to ensure you stay on track.

Common mistakes

❑ Overloading with technical details

- ✗ Spending too much technical details may make your audiences overwhelmed or your presentation running out of time.
- ✓ Simplify complex concepts and break them down into digestible parts. Use analogies or examples that are relatable.

Common mistakes

❑ Not knowing your audiences

- ✗ Failing to tailor the content to the audience's level of technical understanding.
- ✓ Adjust your language and detail level based on the audience's knowledge—avoid over-explaining or under-explaining.

Common mistakes

❑ Reading directly from slides

- ✗ Reading text verbatim from the slides makes the presentation monotonous and disengaging.
- ✓ Use slides as prompts or summaries. Speak more naturally and engage with the audience.

Common mistakes

❑ Lack of practice

- ✗ Rushing through or being unsure of content due to insufficient practice.
- ✓ Rehearse multiple times to ensure smooth delivery, proper timing, and familiarity with the material.

Common mistakes

❑ Skipping over questions or Q&A

- ✗ Ignoring questions, answering dismissively, or running out of time for Q&A.
- ✓ Set specific time for questions and manage it effectively.
- ✓ If unsure of an answer, acknowledge it and offer to follow up later.

References

- ❑ *How to give a good presentation*, <https://www.princeton.edu/~archss/webpdfs08/BaharMartonosi.pdf>
- ❑ Michael Ernst, *How to give a technical presentation (how to give a scientific talk)*, <https://homes.cs.washington.edu/~mernst/advice/giving-talk.html>
- ❑ Ian Parberry, *How to Present a Paper in Theoretical Computer Science: A Speaker's Guide for Students*, <https://ianparberry.com/pubs/speaker.pdf>